

*Blue texts are hyperlinks.

🎓 EDUCATION

University of California, Los Angeles (UCLA) Computer Science Jan. 2025 – Mar. 2026

- Visiting graduate researcher; research: urban embodied navigation, VLA.
- Advisor: Prof. Bolei Zhou.

Zhejiang University College of Control Science and Engineering Sep. 2021 – Present

- Ph.D. candidate; research: embodied navigation, LLM agents, autonomous driving perception.
- Institute of Advanced Perception and Robotics (APRIL Lab).
- Advisor: Prof. Yong Liu.

Zhejiang University of Technology College of Information Engineering (Honors Program) Sep. 2017 – Jun. 2021

- B.Eng.; Provincial Outstanding Graduate; campus awards in speaking/singing; president of the Xianyu Guitar Club.

🔬 RESEARCH

During my Ph.D., I have published **25+ papers** (15 as first/co-first author) with **500+ citations**, including CVPR, ICLR, NeurIPS, ICCV, ICRA, and journals such as TNNLS, TITS, JFR, and RAL. My research started from SLAM and sensor fusion, expanded to autonomous driving perception and foundation-model decision making, and recently focuses on **embodied navigation and urban-scene agents**.

1. Embodied Navigation & VLA

AURA: Multimodal Shared Autonomy for Real-World Urban Navigation [CVPR 2026](#) (First author)

- Propose a multimodal shared-autonomy framework based on a **VLA** architecture for wheeled/quadruped robot navigation in real urban environments. The system fuses language instructions, visual observations, and proprioception to directly output low-level control commands for **end-to-end urban navigation**.
- Design intent understanding and shared-control mechanisms to significantly reduce teleoperation frequency/cost in human-robot collaboration, supporting multimodal inputs (sketch/text/keyboard&mouse).

From Seeing to Experiencing: Scaling Navigation Foundation Models with RL [ICLR 2026](#) (Co-first author)

- Key idea: navigation foundation models learn to *see* from offline videos but lack interactive *experience*. We propose a two-stage paradigm: large-scale video behavior-cloning pretraining for general priors, followed by RL post-training with a **Residual Attention Module (RAM)** that freezes pretrained weights and trains a learnable parallel copy to acquire obstacle avoidance and dynamic interaction without catastrophic forgetting.
- Build NavBench-GS, a closed-loop benchmark based on 3D Gaussian Splatting; RL post-training improves success rate by **21%** in obstacle-rich scenarios; deployed on wheeled, quadruped, and humanoid platforms, demonstrating cross-embodiment transfer.

Open World Sidewalk Occupancy Prediction via Hybrid Training *Submitted to ECCV 2026* (First author)

- Release the **first multi-sensor dataset for urban sidewalk robotics**, and establish a 2D–3D perception benchmark with high-quality human annotations.
- Introduce a hybrid 2D–3D consistency training strategy under scarce 3D labels, leveraging large-scale 2D data to enhance spatiotemporal generalization and to effectively train/evaluate **open-world 3D occupancy prediction**. The model is lightweight with simple inputs, enabling easy deployment on common robot platforms.

2. Foundation-Model-Driven Autonomous Driving

LeapVAD: Cognitive Perception and Dual-Process Thinking for AD [TNNLS 2025](#) (First author)

- A major upgrade of LeapAD. Core innovations: (1) a VLM-finetuned **cognitive perception module** that mimics selective attention and produces semantic/spatial/motion/behavioral reasoning descriptions for critical traffic entities; (2) a ViT-based **Scene Encoder** trained with contrastive learning to produce compact scene tokens, replacing text embeddings for millisecond-level experience retrieval.

- Introduce a **reflection** mechanism for continual improvement from mistakes; GPT-4o performs slow reasoning to curate the memory bank, and a lightweight 1.8B model inherits knowledge for real-time deployment. Achieves pure-vision SOTA in CARLA (DS **88.19%**) and demonstrates cross-domain transfer from CARLA to DriveArena.

LeapAD: Continuously Learning, Adapting, and Improving *NeurIPS 2024* (Co-first author)

- Introduce **dual-process theory** into autonomous driving decision-making, proposing a “slow thinking → fast reaction” distillation paradigm: GPT-4 as System-II performs deep reasoning, accumulates driving experience, and learns from failures; a lightweight 1.8B System-I model inherits capabilities via SFT for real-time deployment (**5×** faster inference).
- Key strengths: **low data demand** (11K labeled vs. 720K–3M in prior work), **continual learning** via closed-loop interaction and memory expansion, and few-shot retrieval with cross-city generalization. CARLA DS **83.11%**, surpassing comparable pure-vision methods.

3. World Models & Driving Simulation

DriveArena: A Closed-loop Generative Simulation Platform *ICCV 2025* (Co-first author)

- Build the **first closed-loop autonomous driving simulation system based on generative models**. Key idea: replace conventional rendering engines with a generative World Dreamer, producing realistic multi-view driving scenes without pre-built assets. Autoregressive generation supports arbitrarily long video synthesis, while a Traffic Manager handles multi-agent interaction and collision detection.
- Highly modular architecture: Driving Agent, Traffic Manager, and World Dreamer communicate via network interfaces and are **independently replaceable**. Compatible with global OpenStreetMap road networks, enabling zero-shot transfer from nuScenes to nuPlan.

Sparse world models and multimodal sensor data generation (with Huawei Noah’s Ark Lab, Project Lead) Oct. 2024 – Present

- **SparseWorld** (submitted to IROS 2026): propose a lightweight world model with sparse scene representation, directly predicting future 3D boxes and map elements; significantly reduces collision risk on nuScenes open-loop and Bench2Drive closed-loop planning.
- **Multimodal scene generation** (accepted to CVPR 2026): build a closed-loop-integratable driving world model from sparse perception elements, advancing a consistent generation chain from Latent → Occ → LiDAR; post-trained on Waymo; developed/validated on Cosmos Transfer 2.5.

4. 3D Perception & Sensor Fusion

LiCROcc: Teach Radar for Accurate Semantic Occupancy Prediction *RAL 2024* (First author)

- Key challenge: radar point clouds are extremely sparse (~2K points), making 3D semantic occupancy prediction difficult. We propose a **teacher–student cross-modal distillation** paradigm: a LiDAR–camera fusion teacher transfers BEV-space knowledge to a radar student via representation distillation and pixel-level distillation.
- First to demonstrate that **sparse radar can perform semantic occupancy prediction**: pure-radar mIoU improves by **44.1%**. Meanwhile, it preserves radar’s robustness in adverse weather/low light, enabling low-cost all-weather perception.

RIDERS: Radar-Infrared Depth Estimation for Robust Sensing *TITS 2024* (Co-first author)

- Core insight: RGB cameras degrade at night/in smoke; LiDAR suffers in rain/fog, while **thermal infrared + 4D mmWave radar** are complementary. We fuse them via Transformer cross-attention to learn radar–pixel association for all-weather dense depth estimation.
- Build the ZJU-Multispectrum dataset (clear/night/fog scenes), outperforming pure-vision solutions under extreme conditions.

RadarCam-Depth: Radar-Camera Fusion for Depth Estimation *ICRA 2024* (Co-first author)

- Key idea: monocular depth lacks absolute scale, while radar provides accurate range but is sparse. Instead of directly fusing heterogeneous features, we use radar as a **scale anchor**: predict relative depth from a monocular network, learn quasi-dense radar–pixel association via Transformer cross-attention, and then refine depth scale.
- Reduce MAE by **25.6%** on nuScenes vs. prior SOTA; reduce MAE by 40.2% on the in-house ZJU-4DRadarCam dataset, supporting the design choice of “scale enhancement” over “feature fusion”.

5. SLAM & Localization

RoLM: Radar on LiDAR Map Localization *ICRA 2023* (First author)

- Problem: radar is all-weather but low-resolution; LiDAR maps are accurate but costly to build. We propose a heterogeneous framework for **online radar localization on offline high-definition LiDAR maps**: use conditional GAN to translate radar BEV into “pseudo-LiDAR” representations, then perform global localization via Monte Carlo particle filtering.
- Validated across 8 days (multi-session) on Oxford Radar RobotCar; achieves **robust all-weather localization** in GNSS-challenged urban canyons.

FMCW Radar on LiDAR Map Localization in Urban Environments

JFR 2024 (First author)

- Journal extension of RoLM. Propose a **Scan Projection Descriptor** to embed radar and LiDAR scans into unified density-map representations for cross-modal place recognition (coarse-to-fine: retrieval then ICP), and systematically summarize localization challenges introduced by FMCW radar scattering characteristics.
- Improve accuracy via echo filtering and observability enhancement; achieve large gains over RoLM on MulRan and Oxford (position RMSE 2.53m, heading 1.83°).

D-GLSNet: Geo-localization with Transformer-based 2D-3D Match Network

RAL 2023 (Co-first author)

- Key idea: traditional geo-localization relies on keypoint detection/matching and is sensitive to viewpoint/appearance changes. We propose a Transformer-based **end-to-end 2D-3D direct matching network** that establishes pixel-point correspondences between satellite imagery and LiDAR point clouds without explicit keypoints, directly regressing 3-DoF pose offsets.
- Design a plug-and-play geo-localization node to provide global constraints for SLAM systems; maintains stable localization under GPS outages (KITTI average RTE **1.43m**).

EXPERIENCE

University of California, Los Angeles (UCLA) Visiting Graduate Researcher 2025 – 2026

- **Urban-scene perception:** focus on sidewalk robotics perception, filling the gap in datasets and evaluation protocols; propose a hybrid training pipeline to mitigate 2D/3D distribution imbalance, covering pseudo-label generation, hybrid training, and high-quality human evaluation.
- **Urban-scene planning:** explore general-purpose navigation foundation models via pretraining and RL post-training; study efficient human-agent interaction paradigms for navigation in the foundation-model era, emphasizing Human-Centric AI.
- **Outputs:** 6 research outcomes in total, including 2 first-author papers (CVPR 2026, ECCV 2026), and 1 co-authored paper (ICLR 2026).

Shanghai AI Laboratory Autonomous Driving Lab, Research Intern

2023 – 2024

- **Foundation-model-driven AD:** explore LLM/VLM for autonomous driving; develop the LeapAD/LeapVAD series for closed-loop driving decision-making and continual learning via dual-process cognitive theory.
- **Driving simulation:** lead development of the closed-loop generative simulation platform DriveArena, integrating multiple vision-based driving agents and supporting diverse scenario generation/evaluation over global road networks.

SKILLS

- **Programming & tools:** C/C++, Python, PyTorch, Git, L^AT_EX, ROS/ROS2, Ceres, Eigen, g2o, GTSAM
- **Foundation models & embodied stacks:** GR00T N1, LLaVA, Qwen-VL, InternVL, CityWalker, Isaac Sim
- **Autonomous driving stacks:** Magic Drive, MapTR, BEVFusion, UniAD, CARLA

HONORS & AWARDS

- Outstanding Communist Party Member, Zhejiang University 2025
- ECCV 2024 Workshop (Anomaly Scenarios in Autonomous Driving): **Second Prize** 2024
- Outstanding Graduate Student, Zhejiang University 2017–2021
- Provincial Outstanding Graduate 2017–2021
- Provincial Government Scholarship (3 consecutive years) 2017–2021
- Special Prize, Canal Cup (university competition) 2020
- National Undergraduate Intelligent Vehicle Competition (Beacon Group): **First Prize** 2020
- National Undergraduate Mathematical Contest in Modeling: **First Prize** 2019

SELECTED PUBLICATIONS (SEE MORE ON [GOOGLE SCHOLAR](#))

1. **Yukai Ma**, Honglin He, et al., AURA: Multimodal Shared Autonomy for Real-World Urban Navigation, *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
2. **Yukai Ma**, Tiantian Wei, et al., LeapVAD: A Leap in Autonomous Driving via Cognitive Perception and Dual-Process Thinking, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2025.
3. **Yukai Ma***, Jianbiao Mei*, et al., LiCROcc: Teach Radar for Accurate Semantic Occupancy Prediction using LiDAR and Camera, *IEEE Robotics and Automation Letters (RAL)*, 2024.
4. **Yukai Ma**, et al., FMCW Radar on LiDAR Map Localization in Structural Urban Environments, *Journal of Field Robotics (JFR)*, 2024.
5. **Yukai Ma**, et al., RoLM: Radar on LiDAR Map Localization, *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

6. Honglin He*, **Yukai Ma***, et al., From Seeing to Experiencing: Scaling Navigation Foundation Models with Reinforcement Learning, *International Conference on Learning Representations (ICLR)*, 2026.
7. Xuemeng Yang*, Licheng Wen*, Tiantian Wei*, **Yukai Ma***, Jianbiao Mei*, et al., DriveArena: A Closed-loop Generative Simulation Platform for Autonomous Driving, *International Conference on Computer Vision (ICCV)*, 2025.
8. Jianbiao Mei*, **Yukai Ma***, et al., Continuously Learning, Adapting, and Improving: A Dual-Process Approach to Autonomous Driving, *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
9. Han Li* **Yukai Ma***, et al., RIDERS: Radar-Infrared Depth Estimation for Robust Sensing, *IEEE Transactions on Intelligent Transportation Systems (TITS)*, 2024.
10. Han Li* **Yukai Ma***, et al., RadarCam-Depth: Radar-Camera Fusion for Deep Metric Depth Estimation, *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.